Introduction to Regression-II

Simple Linear Regression

Multiple Linear Regression

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TV ads X	Cars sold Y
1	14
3	24
2	18
1	17
3	27
$\sum X = 10$	$\sum Y = 100$
$\bar{X} = 2$	$\bar{Y} = 20$

Estimated simple linear regression equation :-

$$\hat{y} = b_0 + b_1 x$$

Graph is called estimated regression line.

b_0 is the y intercept of the line

b_1 is the slope of the line

$\hat{y}$ is the estimated value of y for a given value x.

Least Squares Criteria:

$$\min \sum (y_i - \hat{y}_i)^2$$

observed value

estimated value

- One ... slope of the estimated regression equation:

$$b_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

x_i independent variable

y_i dependent variable

$\bar{x}$ $\bar{y}$ = mean values for x & y.

$$b_0 = \bar{y} - b_1 \bar{x}$$

Y-intercept for the estimated regression equation.

Soln:-

b_1 = slope

$$b_1 = (1-2)(14-20) + (3-2)(24-20) + (2-2)(18-20) + (1-2)(17-20) + (3-2)(27-20)$$

$$= (-1)(-6) + (1)(4) + 0 + (-1)(-3) + (1)(7)$$

$$= 6 + 4 + 0 + 3 + 7$$

$$= 20$$

$$(1-2)^2 + (3-2)^2 + (2-2)^2 + (1-2)^2 + (3-2)^2$$

$$(-1)^2 + (1)^2 + 0 + (-1)^2 + (1)^2$$

$$1 + 1 + 0 + 1 + 1 = 4$$

$$b_1 = \frac{20}{4} = 5$$

- One intercept for the estimated regression equation.

$$b_0 = \bar{y} - b_1\bar{x}$$
$$b_0 = 20 - 5(2)$$
$$b_0 = 10$$

Estimated regression equation:-

$$\boxed{\hat{Y} = 10 + 5X} \quad \text{———} \quad (2)$$

Coefficient of Determination

$$SST = SSR + SSE$$

$$\sum (y_i - \bar{y})^2 = \sum (\hat{y}_i - \bar{y})^2 + \sum (y_i - \hat{y})^2$$

SST = total [sum of square (total)]

SSR = sum of squares due to regression

SSE = sum of squares due to error

Coefficient of determination r^2

$$r^2 = \frac{SSR}{SST}$$

$$r^2 = \frac{SSR}{SST} = \frac{100}{114} = .8772$$

$\therefore$ Regression relationship is strong i.e. 87.72% of the variability in the number of car sold can be explained by linear relationship.

$$SST = \sum (y_i - \bar{y})^2$$

$$= (14-20)^2 + (24-20)^2 + (18-20)^2 + (17-20)^2 + (27-20)^2$$

$$= (-6)^2 + (4)^2 + (-2)^2 + (-3)^2 + (7)^2$$

$$= 36 + 16 + 4 + 9 + 49$$

$$= \boxed{114}$$

x	y
1	14
3	24
2	18
1	17
3	27
	$\bar{y} = 20$

$$SSR = \sum (\hat{y}_i - \bar{y})^2$$

$$= (15-20)^2 + (25-20)^2 + (20-20)^2 + (15-20)^2 + (25-20)^2$$

$$= (-5)^2 + (5)^2 + 0 + (-5)^2 + (5)^2$$

$$= 25 + 25 + 0 + 25 + 25$$

$$= \boxed{100}$$

① $\hat{y} = 10 + 5(x)$
$= 15$

② $\hat{y} = 10 + 5(2)$
$= 20$

③ $\hat{y} = 10 + 5(3)$
$= 25$

$$r^2 = \frac{100}{114} = .8772$$

$$\boxed{\simeq 87.72}$$

Sample correlation Cofficent :-

$$r_{xy} = (\text{Sign of } b_1)\sqrt{r^2}$$

$$= +\sqrt{0.8772}$$

$$r_{xy} = +.9366$$

Multiple Regression

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- It is a study of more than two variables
- One variable is dependent variable (X_1) & other variables (X_2 & X_3) are independent variables.
- It is used for prediction
- we predict the change in Dependent variable according to change in Independent variables.

Example

Situation	Independent var 1 (Price)	Independent var 2 (Advertisement)	Dependent variable (Sales)
Case 1	50	2,00,000	5,00,000
2	40	2,20,000	6,00,000
3	60	1,80,000	4,00,000
4	55	2,10,000	?
5	?	2,00,000	7,00,000
6	40	?	65,00,00

Methods for Multiple Regression

- Least Square method / (Direct method)
- Least Square method using Mean ($\bar{X}$) / (Short-cut Meth

Multiple Regression Equation of X_1 on X_2 & X_3

Line

$$X_1 = a_{1.23} + b_{12.3} X_2 + b_{13.2} X_3$$

where

X_1 = Dependent variable

X_2 & X_3 = Independent variables

Σ = Sigma

$b_{12.3}$ & $b_{13.2}$ = Partial regression Coefficients

Using Least Square method

For X_1:- The value of constants $a_{1.23}$, $b_{12.3}$ & $b_{13.2}$ are obtained by solving the following three Normal equations:-

$$\sum X_1 = N.a_{1.23} + b_{12.3} \sum X_2 + b_{13.2} \sum X_3 \quad \text{(i)}$$

$$\sum X_1 X_2 = a_{1.23} \sum X_2 + b_{12.3} \sum (X_2)^2 + b_{13.2} \sum X_2 X_3 \quad \text{(ii)}$$

$$\sum X_1 X_3 = a_{1.23} \sum X_3 + b_{12.3} \sum X_2 X_3 + b_{13.2} \sum (X_3)^2 \quad \text{(iii)}$$

For X_2:-

$$X_2 = a_{2.13} + b_{21.3} X_1 + b_{23.1} X_3$$

X_2 = Dep var X_1 & X_3 = Ind. var $b_{21.3}$ & $b_{23.1}$ = Partial Reg Coeff

$$\sum X_2 = N \cdot a_{2.13} + b_{21.3} \sum X_1 + b_{23.1} \sum X_3$$

$$\sum X_2 X_1 = a_{2.13} \sum X_1 + b_{21.3} \sum (X_1)^2 + b_{23.1} \sum X_3 X_1$$

$$\sum X_2 X_3 = a_{2.13} \sum X_3 + b_{21.3} \sum X_1 X_3 + b_{23.1} \sum (X_3)^2$$

Similarly, third equation will be

$$X_3 = a_{3.12} + b_{31.2} X_1 + b_{32.1} X_2$$ line.

X_3 = Dependent variable

X_1 & X_2 = Independent variables

$b_{31.2}$ & $b_{32.1}$ = Partial reg co efficients

$$\sum X_3 = N \cdot a_{3.12} + b_{31.2} \sum X_1 + b_{32.1} \sum X_2$$

$$\sum X_3 X_1 = a_{3.12} \sum X_1 + b_{31.2} \sum (X_1)^2 + b_{32.1} \sum X_2 X_1$$

$$\sum X_3 X_2 = a_{3.12} \sum X_2 + b_{31.2} \sum X_1 X_2 + b_{32.1} \sum (X_2)^2$$

Question 1:- Calculate Multiple Regression equation of X_1 on X_2 & X_3

X_1	4	6	7	9	13	15
X_2	15	12	8	6	4	3
X_3	30	24	20	14	10	4

Solution:- The regression equation of X_1 on X_2 & X_3

$$X_1 = a_{1.23} + b_{12.3} X_2 + b_{13.2} X_3$$

Three normal equations:-

$$\Sigma X_1 = N \cdot a_{1.23} + b_{12.3} \Sigma X_2 + b_{13.2} \Sigma X_3 \quad \text{--(i)}$$

$$\Sigma X_1 X_2 = a_{1.23} \Sigma X_2 + b_{12.3} \Sigma (X_2)^2 + b_{13.2} \Sigma X_2 X_3 \quad \text{--(ii)}$$

$$\Sigma X_1 X_3 = a_{1.23} \Sigma X_3 + b_{12.3} \Sigma X_2 X_3 + b_{13.2} \Sigma (X_3)^2 \quad \text{--(iii)}$$

X_1	X_2	X_3	X_1X_2	X_1X_3	X_2X_3	X_2^2	X_3^2
4	15	30	60	120	450	225	900
6	12	24	72	144	288	144	576
7	8	20	56	140	160	64	400
9	6	14	54	126	84	36	196
13	4	10	52	130	40	16	100
15	3	4	45	60	12	9	16
$\Sigma X_1 = 54$	$\Sigma X_2 = 48$	$\Sigma X_3 = 102$	$\Sigma X_1X_2 = 339$	$\Sigma X_1X_3 = 720$	$\Sigma X_2X_3 = 1034$	$\Sigma X_2^2 = 494$	$\Sigma X_3^2 = 2188$

Fit these values in Regression equations

ΣX_1	ΣX_2	ΣX_3	ΣX_1X_2	ΣX_1X_3	ΣX_2X_3	$\Sigma (X_2)^2$	$\Sigma ($
54	48	102	339	720	1034	794	2188

Substituting the values in the above three equations:—

N = 6 (instances / Bonds)

$$54 = 6a_{1.23} + 48b_{12.3} + 102b_{13.2} \quad \text{—— (iv)}$$

$$339 = 48a_{1.23} + 494b_{12.3} + 1034b_{13.2} \quad \text{—— (v)}$$

$$720 = 102a_{1.23} + 1034b_{12.3} + 2188b_{13.2} \quad \text{—— (vi)}$$

Equal eq 1 & 2:— multiplying 8:—

Now, try to equate any two equations.

$$339 = 48a_{1.23} + 494b_{12.3} + 1034b_{13.2} \quad \text{—— (vii)}$$

Multiplying 8, we get.

$$432 = 48a_{1.23} + 384b_{12.3} + 816b_{13.2} \quad \text{—— (viii)}$$

Subtracting (viii) from (vii):

$$-93 = 0 + 110b_{12.3} + 218b_{13.2} \quad \text{—— (ix)}.$$

Now take eq (iv) & (vi)

$$54 = 6a_{1.23} + 48b_{12.3} + 102b_{13.2} \quad \text{——— (iv)}$$

$$720 = 102a_{1.23} + 1034b_{12.3} + 2188b_{13.2} \quad \text{——— (vi)}$$

Multiply (ix) by 17, we get:

$$918 = 102a_{1.23} + 816b_{12.3} + 1734b_{13.2} \quad \text{——— (x)}$$

Now subtract x from vi

$$-198 = 0 + 218b_{12.3} + 454b_{13.2} \quad \text{——— (xi)}$$

$$-93 = 0 + 110b_{12.3} + 218b_{13.2} \quad \text{——— (xii)}$$

Multiplying (xii) by 109, we get

$$-10137 = 11990b_{12.3} + 23762b_{13.2} \quad \text{——— (x}$$

multiply (xi) 55, we get:

$$-10890 = 11990b_{12.3} + 24970b_{13.2} \quad \text{——— (xiv)}$$

Subtracting (xiv) from (xiii)

$$-10137 = 11990\, b_{12.3} + 23762\, b_{13.2} \quad (xi)$$

$$-10890 = 11990\, b_{12.3} + 24970\, b_{13.2} \quad (xii)$$

$$753 = 0 - 1208\, b_{13.2}$$

$$b_{13.2} = -\frac{753}{1208} = -0.623$$

Now Substituting the value of $b_{13.2}$ in equation (xii)

$$-93 = 0 + 110\, b_{12.3} + 218\, b_{13.2} \quad \text{—— (xii)}$$

$$-93 = 0 + 110\, b_{12.3} + 218(-0.623)$$

$$135.814 - 93 = 110\, b_{12.3}$$

$$b_{12.3} = \frac{42.814}{110} = 0.389$$

Substituting the values of $b_{12.3}$ & $b_{13.2}$ in (iv) we have:-

$$54 = 6a_{1.23} + 48\, b_{12.3} + 102\, b_{13.2} \quad \text{—— (iv)}$$

$$54 = 6a_{1.23} + 48(0.389) + 102(-0.623)$$

$$54 = 6a_{1.23} + 18.672 - 63.546$$

$$6a_{1.23} = 54 - 18.672 + 63.546$$

$$6a_{1.23} = 98.874$$

$$a_{1.23} = \frac{98.874}{6} = 16.479$$

Hence the required equation is:

$$X_1 = a_{1.23} + b_{12.3} X_2 + b_{13.2} X_3$$

$$X_1 = 16.479 + 0.389 X_2 - 0.623 X_3$$

Least square Method using Mean ($\bar{X}$)
(short cut method) *(For more data, complicated numbers)

Multiple Regression equation of X_1 on X_2 & X_3 in deviation form is given by:-

$$X_1 - \bar{X}_1 = b_{12.3}(X_2 - \bar{X}_2) + b_{13.2}(X_3 - \bar{X}_3)$$

where

$$x_1 = (X_1 - \bar{X}_1), \quad x_2 = (X_2 - \bar{X}_2), \quad x_3 = (X_3 - \bar{X}_3)$$

or

$$\boxed{x_1 = b_{12.3} x_2 + b_{13.2} x_3}$$ Regression line

$b_{12.3}$ & $b_{13.2}$ = Partial Regression Co-efficients

The values of the Partial regression co-efficients can be obtained by solving the following two normal equations

$$\Sigma x_1 x_2 = b_{12.3} \Sigma x_2^2 + b_{13.2} \Sigma x_3 x_2$$

$$\Sigma x_1 x_3 = b_{12.3} \Sigma x_2 x_3 + b_{13.2} \Sigma x_3^2$$

Further solved:- we have

$$b_{12.3} = \frac{(\Sigma x_1 x_2)(\Sigma x_3^2) - (\Sigma x_1 x_3)(\Sigma x_2 x_3)}{(\Sigma x_2^2)(\Sigma x_3^2) - (\Sigma x_2 x_3)^2} \quad (1)$$

$$b_{13.2} = \frac{(\Sigma x_1 x_3)(\Sigma x_2^2) - (\Sigma x_1 x_2)(\Sigma x_3 x_2)}{(\Sigma x_3^2)(\Sigma x_2^2) - (\Sigma x_3 x_2)^2} \quad (2)$$

Multiple Regression equation of X_2 on X_1 & X_3 in deviation form is given by:-

$$X_2 - \bar{X}_2 = b_{21.3}(X_1 - \bar{X}_1) + b_{23.1}(X_3 - \bar{X}_3)$$

where, $x_1 = (X_1 - \bar{X}_1)$, $x_2 = (X_2 - \bar{X}_2)$, $x_3 = (X_3 - \bar{X}_3)$

or $$x_2 = b_{21.3} x_1 + b_{23.1} x_3$$

$b_{21.3}$ & $b_{23.1}$ = Partial Regression Co-efficients

The values of the partial Regression coefficients can be obtained by solving the following two normal equations:-

$$\Sigma x_2 x_1 = b_{23.1} \Sigma x_1^2 + b_{23.1} \Sigma x_1 x_3$$

$$\Sigma x_2 x_3 = b_{21.3} \Sigma x_1 x_3 + b_{23.1} \Sigma x_3^2$$

Further solved, we have:-

$$b_{21.3} = \frac{(\Sigma x_2 x_1)(\Sigma x_3^2) - (\Sigma x_2 x_3)(\Sigma x_1 x_3)}{(\Sigma x_1^2)(\Sigma x_3^2) - (\Sigma x_1 x_3)^2}$$

$$b_{23.1} = \frac{(\Sigma x_2 x_3)(\Sigma x_1^2) - (\Sigma x_2 x_1)(\Sigma x_3 x_1)}{(\Sigma x_3^2)(\Sigma x_1^2) - (\Sigma x_3 x_1)^2}$$

Multiple Regression equation of X_3 on X_1 & X_2 in deviation form is given by:

$$X_3 - \bar{X}_3 = b_{31.2}(X_1 - \bar{X}_1) + b_{32.1}(X_2 - \bar{X}_2)$$

where

$$x_1 = (X_1 - \bar{X}_1), \quad x_2 = (X_2 - \bar{X}_2), \quad x_3 = (X_3 - \bar{X}_3)$$

or $x_3 = b_{31.2}\, x_1 + b_{32.1}\, x_2$

$b_{31.2}$ & $b_{32.1}$ = Partial Regression Coefficients

The values of the partial regression coefficients can be obtained by solving the two normal equations

$$\sum x_1 x_3 = b_{31.2} \sum x_1^2 + b_{32.1} \sum x_1 x_2$$

$$\sum x_2 x_3 = b_{31.2} \sum x_1 x_2 + b_{32.1} \sum x_2^2$$

Further solved, we have:

$$b_{31.2} = \frac{\left(\sum x_3 x_1\right)\left(\sum x_2^2\right) - \left(\sum x_3 x_2\right)\left(\sum x_1 x_2\right)}{\left(\sum x_1^2\right)\left(\sum x_2^2\right) - \left(\sum x_1 x_2\right)^2}$$

$$b_{32.1} = \frac{\left(\sum x_3 x_2\right)\left(\sum x_1^2\right) - \left(\sum x_3 x_1\right)\left(\sum x_2 x_1\right)}{\left(\sum x_2^2\right)\left(\sum x_1^2\right) - \left(\sum x_2 x_1\right)^2}$$

Re/ Find the least square regression of X_3 on X_1 & X_2 using actual mean ($\bar{X}$) method, also estimate X_3, when $X_1 = 10$ & $X_2 = 6$.

X_1	3	5	6	8	12	14
X_2	16	10	7	4	3	2
X_3	90	72	54	42	30	12

X_1	x_1	x_1^2	X_2	x_2	x_2^2	X_3	x_3	x_3^2	x_1x_2	x_1x_3	x_2x_3
3	−5	25	16	9	81	90	40	1600	−45	−200	360
5	−3	9	10	3	9	72	22	484	−9	−66	66
6	−2	4	7	0	0	54	4	16	0	−8	0
8	0	0	4	−3	9	42	−8	64	0	0	24
12	4	16	3	−4	16	30	−20	400	−16	−80	80
14	6	36	2	−5	25	12	−38	1444	−30	−228	190
$\Sigma X_1 = 48$	$\Sigma x_1 = 0$	$\Sigma x_1^2 = 90$	$\Sigma X_2 = 42$	$\Sigma x_2 = 0$	$\Sigma x_2^2 = 140$	$\Sigma X_3 = 300$	$\Sigma x_3 = 0$	$\Sigma x_3^2 = 4008$	$\Sigma x_1x_2 = -100$	$\Sigma x_1x_3 = -582$	$\Sigma x_2x_3 = 720$

$x_1 = (X_1 - 8)$, $x_2 = (X_2 - 7)$, $x_3 = (X_3 - 50)$

$\bar{X}_1 = \frac{48}{6} = 8$, $\bar{X}_2 = \frac{42}{6} = 7$, $\bar{X}_3 = \frac{300}{6} = 50$

Regression equation of X_3 on X_1 & X_2 is:

$$X_3 - \bar{X}_3 = b_{31.2}(X_1 - \bar{X}_1) + b_{32.1}(X_2 - \bar{X}_2) \quad \text{——— (a)}$$

$$b_{31.2} = \frac{(\Sigma x_3 x_1)(\Sigma x_2^2) - (\Sigma x_3 x_2)(\Sigma x_1 x_2)}{(\Sigma x_1^2)(\Sigma x_2^2) - (\Sigma x_1 x_2)^2}$$

$$b_{31.2} = \frac{(-582)(140) - (720)(-100)}{(90)(140) - (-100)^2} = \frac{-81480 + 72000}{12600 - 10000}$$

$$= -\frac{9480}{2600} = \boxed{-3.646}$$

$$b_{32.1} = \frac{(\Sigma x_3 x_2)(\Sigma x_1^2) - (\Sigma x_3 x_1)(\Sigma x_2 x_1)}{(\Sigma x_2^2)(\Sigma x_1^2) - (\Sigma x_2 x_1)^2}$$

$$= \frac{(720)(90) - (-582)(-100)}{(90)(140) - (-100)^2}$$

$$= \frac{64800 - 58200}{12600 - 10000} = \frac{6600}{2600} = \boxed{2.538}$$

$$X_3 - 50 = -3.646(X_1 - 8) + 2.538(X_2 - 7)$$

$$X_3 - 50 = -3.646X_1 + 29.168 + 2.538X_2 - 17.766$$

$$\boxed{X_3 = -3.646X_1 + 2.538X_2 + 61.402}$$

<u>Prediction:-</u> $X_1 = 10$ & $X_2 = 6$, so $X_3 = ?$

$$X_3 = -3.646(10) + 2.538(6) + 61.402$$

$$= -36.46 + 15.228 + 61.402$$

$$= 40.17 \simeq 40$$

$$\boxed{X_3 = 40}$$ <u>Ans</u>

Standard Error of Estimate or

Reliability of Estimate

- It measures the reliability of estimate

x_3